

Daoyuan Zheng

Ph.D. Candidate at Wuhan University

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Research Interests

My research lies at the intersection of intelligent remote sensing interpretation and digital cultural heritage. One direction focuses on semi- and weakly supervised methods for building extraction, road extraction, and change detection. Another direction concerns deep learning-based archaeological relic identification and cultural heritage discovery. This work also involves the construction of a large-scale relic segmentation dataset for Great Wall linear archaeological traces.

Education

Ph.D. Candidate in Photogrammetry and Remote Sensing Wuhan University Adviser: Prof. Qingwu Hu & Prof. Shaohua Wang	09/2023 – Present
M.E. in Computer Science and Technology China University of Geosciences (Wuhan) Adviser: Assoc. Prof. Fang Fang	09/2020 – 06/2023
B.E. in Software engineering Wuhan Textile University	09/2016 – 06/2020

Projects

National Key R&D Program of China, Grant No. 2024YFB3908900 <i>Student Researcher - Dynamic Monitoring and Risk Assessment of Large-Scale Linear Cultural Heritage</i>	2025 – Present
National Key R&D Program of China, Grant No. 2023YFF0906203 <i>Student Researcher - Key Technologies on Digital Fingerprint Extraction, Authentication and Intelligent Identity Management of Museum Collections</i>	2023 – Present
National Key R&D Program of China, Grant No. 2021YFB2600401 <i>Student Researcher - High-precision Intelligent Surveying and Mapping Technology Integrated with Beidou Positioning and Space-Air-Ground Observation</i>	2023 – 2024
Key Special Project of Ministry of Natural Resources <i>Student Researcher - Fine-Grained Land Use Classification Integrating Multi-Modal Data</i>	12/2021 – 06/2023
Tencent Cloud Intelligent R&D Company <i>Student Intern, CSIG - Research on Community Detection of Tencent Meeting Participants</i>	06/2022 – 08/2022
Open Fund Project of National Center for Earth Observation Science <i>Student Researcher - Construction of Building Instance Segmentation Dataset from Remote Sensing Imagery</i>	09/2020 – 08/2021

Academic Research

Semi-Supervised Great Wall Relic Trace Extraction via Scene-level Adapter and Relic-oriented Consistency Learning <i>IEEE Transactions on Geoscience and Remote Sensing (Submitted to Journal)</i>	2026
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GWSegNet: A Linear-aware Segmentation Network for Great Wall Trace Identification in High-Resolution Remote Sensing Imagery <i>Engineering Applications of Artificial Intelligence (Under Review)</i> Daoyuan Zheng , Qingwu Hu, Haixia Feng, Tiancheng Liu, Ziyuan Feng, Yinhuo Lan, Rui Xu, Mingyao Ai, Pengcheng Zhao, Zhipeng Chen, Shaohua Wang	2026
Terrain-aware Fusion and Connectivity-aware Learning Network for Great Wall Relic Extraction from Remote Sensing Imagery and Terrain Data <i>IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (Under Review)</i> Daoyuan Zheng , Zhaohui Guo, Tiancheng Liu, Haixia Feng, Jiazheng Liang, Mingyao Ai, Pengcheng Zhao, Shaohua Wang, Zhipeng Chen and Qingwu Hu	2026

First-Author Publications

Automatic detection and site-scale vegetation shift analysis for individual heritage buildings along the Great Wall <i>npj Heritage Science</i> Daoyuan Zheng , Shaohua Wang, Haixia Feng, Tiancheng Liu, Yizhou Lan, Shunli Wang, Xujie Zhang, Mingyao Ai, Pengcheng Zhao, Jiayuan Li, Qingwu Hu	2026
CSNet: Change selection of activations and pseudomasks for image-level weakly supervised change detection <i>IEEE Transactions on Geoscience and Remote Sensing</i> Daoyuan Zheng , Shaohua Wang, Haixia Feng, Rui Xu, Shunli Wang, Mingyao Ai, Pengcheng Zhao, Qingwu Hu	2025
A dataset of multi-source typical land cover classification in the Field Park in Wuhan (2020) <i>China Scientific Data</i> Daoyuan Zheng , Yizhou Lan, Zhongqiao Chen, Ying Zhang, Mingyao Ai, Pengcheng Zhao, Qingwu Hu	2024
Weakly supervised building extraction from high-resolution remote sensing images based on building-aware clustering and activation refinement network <i>IEEE Transactions on Geoscience and Remote Sensing</i> Daoyuan Zheng , Shaohua Wang, Haixia Feng, Shunli Wang, Mingyao Ai, Pengcheng Zhao, Jiayuan Li, Qingwu Hu	2024
Utilizing bounding box annotations for weakly supervised building extraction from remote sensing images <i>IEEE Transactions on Geoscience and Remote Sensing</i> Daoyuan Zheng , Shengwen Li, Fang Fang, Jiahui Zhang, Yuting Feng, Bo Wan, Yuanyuan Liu	2023
Semi-Supervised Building Detection from High-Resolution Remote Sensing Imagery <i>Sustainability</i> Daoyuan Zheng , Jianing Kang, Kaishun Wu, Yuting Feng, Han Guo, Xiaoyun Zheng, Shengwen Li, Fang Fang	2023

Collaborative Publications

Improved pseudomasks generation for weakly supervised building extraction from high-resolution remote sensing imagery <i>IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing</i> Fang Fang, Daoyuan Zheng , Shengwen Li, Yuanyuan Liu, Linyun Zeng, Jiahui Zhang, Bo Wan	2022
Revealing latent linear structures in murals via joint analysis of hyperspectral and visible images <i>npj Heritage Science</i> Yuxiang Xu, Qingwu Hu, Pengcheng Zhao, Mingyao Ai, Shaohua Wang, Jiayuan Li, Daoyuan Zheng , Haixia Feng	2026

A Self-Prompt Calibration Network Based on Segment Anything Model 2 for High-Resolution Remote Sensing Image Segmentation <i>IEEE Geoscience and Remote Sensing Letters</i> Yizhou Lan, Daoyuan Zheng , Xinge Zhao, Ke Shang, Feizhou Zhang	2025
A non-overlapping image stitching method for reconstruction of page in ancient Chinese books <i>Computer Vision and Image Understanding</i> Yizhou Lan, Daoyuan Zheng , Qingwu Hu, Shaohua Wang, Shunli Wang, Tong Yue, Jiayuan Li	2025
Automatic generation of Chinese mural line drawings via enhanced edge detection and CycleGAN-based denoising <i>npj Heritage Science</i> Haixia Feng, Qingwu Hu, Pengcheng Zhao, Daoyuan Zheng , Mingyao Ai, Siliang Chen, Xiyu Hu	2025
CLC2SIE: Cross-level consistency constraint for semi-supervised building instance extraction from high-resolution remote sensing imagery <i>IEEE Transactions on Geoscience and Remote Sensing</i> Rui Xu, Jun Pan, Fang Fang, Daoyuan Zheng , Shengwen Li, Yang Yang	2025
Geothermal detection study using remote sensing data by combining machine learning and deep learning: A case study of Huanggang City <i>Geothermics</i> Haixia Feng, Qingwu Hu, Pengcheng Zhao, Mingyao Ai, Shaohua Wang, Daoyuan Zheng	2025
SFANet: A Ground Object Spectral Feature Awareness Network for Multimodal Remote Sensing Image Semantic Segmentation <i>Remote Sensing</i> Yizhou Lan, Daoyuan Zheng , Yingjun Zheng, Feizhou Zhang, Zhuodong Xu, Ke Shang, Zeyu Wan	2025
FTransDeepLab: Multimodal fusion transformer-based DeepLabv3+ for remote sensing semantic segmentation <i>IEEE Transactions on Geoscience and Remote Sensing</i> Haixia Feng, Qingwu Hu, Pengcheng Zhao, Shunli Wang, Mingyao Ai, Daoyuan Zheng , Tiancheng Liu	2025
AMIANet: Asymmetric multimodal interactive augmentation network for semantic segmentation of remote sensing imagery <i>IEEE Transactions on Geoscience and Remote Sensing</i> Tiancheng Liu, Qingwu Hu, Wenlei Fan, Haixia Feng, Daoyuan Zheng	2024
Spatial Extent-Aware Multimodal Fusion Method for Measuring Urban Socioeconomic Status <i>IEEE International Geoscience and Remote Sensing Symposium</i> Fang Fang, Jiahui Zhang, Shengwen Li, Daoyuan Zheng , Linyun Zeng, Bo Wan	2023
DSM-assisted unsupervised domain adaptive network for semantic segmentation of remote sensing imagery <i>IEEE Transactions on Geoscience and Remote Sensing</i> Shunping Zhou, Yuting Feng, Shengwen Li, Daoyuan Zheng , Fang Fang, Yuanyuan Liu, Bo Wan	2023
Spatial context-aware method for urban land use classification using street view images <i>ISPRS Journal of Photogrammetry and Remote Sensing</i> Fang Fang, Linyun Zeng, Shengwen Li, Daoyuan Zheng , Jiahui Zhang, Yuanyuan Liu, Bo Wan	2022

A coarse-to-fine contour optimization network for extracting building instances from high-resolution remote sensing imagery

2021

Remote Sensing

Fang Fang, Kaishun Wu, Yuanyuan Liu, Shengwen Li, Bo Wan, Yanling Chen, **Daoyuan Zheng**

Honors and Awards

1st Global Digital Intelligence Education Innovation Competition (Gold Medal) Topic: Precise Vector Mapping of Buildings and Roads from UAV Aerial Imagery	11/2024
Academic Scholarship of Wuhan University (First-Class)	10/2024
Outstanding Graduate of China University of Geosciences (Wuhan)	06/2023
Academic Scholarship of China University of Geosciences (Wuhan) (First Class)	10/2021
Academic Scholarship of China University of Geosciences (Wuhan) (First Class)	10/2020
Outstanding Graduate of Wuhan Textile University	06/2020
Academic Scholarship of Wuhan Textile University (Second-Class)	10/2018
Academic Scholarship of Wuhan Textile University (First-Class)	10/2017

Academic Presentations

A Deep Learning Framework for Intelligent Detection of Individual Great Wall Buildings and Vegetation Change Monitoring <i>Wuhan University, Digital Archaeology Annual Conference 2025</i>	12/2025
Weakly Supervised Building Extraction from High-Resolution Remote Sensing Imagery <i>Wuhan University, 9th Zhizhuo Doctoral Forum</i>	12/2024
Weakly Supervised Building Extraction from High-Resolution Remote Sensing Imagery <i>China University of Geosciences (Wuhan), 32nd Graduate Academic Paper Symposium</i>	10/2021

Academic Service

Journal Reviewer

IEEE Transactions on Geoscience and Remote Sensing (TGRS)
International Journal of Applied Earth Observation and Geoinformation (JAG)
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS)
IEEE Geoscience and Remote Sensing Letters (GRSL)